



Alkhidmat Foundation Pakistan

Summer Internship Program 2026

GEN Z Entrepreneurship Initiative

Community Action & Impact Project

Idea Name: NeuroNova

Department: Volunteer Management

Submitted by: Muhammad Taqui

muhammادتaquiachat@gmail.com

Submission Date: August 22, 2026

This report is submitted as the Community Action & Impact Project deliverable of the Alkhidmat Summer Internship Program 2026, presenting NeuroNova as an entrepreneurship concept in response to the GEN Z Entrepreneurship Initiative brief.

1 Project overview

Idea name: NeuroNova. **Format:** an open-source, project-based curriculum in applied artificial intelligence and data science, hosted publicly on GitHub. The idea responds directly to a real, personal starting point: as a final-year AI student based in a smaller Punjab city rather than a major tech hub, structured, hands-on AI/DS training has been consistently harder to access locally than the underlying material itself. NeuroNova turns that observation into a concept aimed at other learners in the same position.

2 Problem identified

Pakistan is producing IT graduates faster than it is producing applied AI skills, and structured hands-on training remains concentrated in a handful of major cities. Over 75,000 IT graduates enter Pakistan's job market every year, yet fewer than 10 percent of active IT professionals have applied AI skills, and roughly 20 percent of engineering and IT graduates meet industry skill standards on graduation.^{1 2}

The access side of this problem is geographic as much as it is educational. Cohort-based training programs typically enrol on physical campuses in Karachi and Lahore. A recent national skills test found that many students had no organised study materials at all going into the assessment.³ Being based in a smaller Punjab city rather than a major hub is the everyday, lived version of this gap.

3 Basic market and competitor research

Three existing programs address adjacent parts of this problem, but each leaves a specific gap that NeuroNova is positioned to fill. Table 1 summarises the comparison.

Table 1: Competitor comparison

Program	Scale	Strength	Where the gap remains
DigiSkills.pk	4.5 million+ enrolments since 2018 (Ignite / Virtual University)	Free, fully remote, explicitly geography-agnostic	Broad digital and freelancing skills, not a structured applied AI/DS path
Bano Qabil	25+ IT courses, now including AI and Data Analytics (Alkhidmat)	Alkhidmat's own program, strong brand trust	Cohort enrolment is campus-based (Karachi/Lahore), with an aptitude test, interview, and refundable deposit
Google Career Certificates	100,000 scholarships announced, Punjab, 2026	Well-funded, recognised credential	Distributed through university partnerships; access depends on the institutional channel

¹Business Recorder, "Pakistan's AI moment," June 2026, citing an Access Partnership estimate of a Rs2.8 trillion GDP upside from closing the digital skills gap.

²Daily Parliament Times, "The Skills Gap Pakistan Cannot Ignore," World Youth Skills Day coverage, July 2026.

³Pakistan Today, coverage of the HEC National Skills Competition Test (NSCT) 2026, which tested roughly 33,000 IT students nationwide.

Sources: digiskills.pk; banoqabil.pk (registration and course pages); Superior Colleges, “Biggest Gaps in College Education System 2026” (Google–Punjab HED partnership coverage).⁴

4 Proposed solution

NeuroNova is an open-source, GitHub-hosted curriculum at bachelor’s level, built around real projects rather than video lectures, and free to start from anywhere with an internet connection. Four properties define it:

- **Project-based.** Learners build real, demonstrable work rather than only watching lectures or collecting certificates.
- **Location-agnostic.** It runs entirely online, with no campus enrolment, aptitude test, or deposit required.
- **Portfolio-first.** Work lives on GitHub from the first lesson, building a track record an employer can verify directly.
- **Free and open.** It is openly licensed and free to use, positioned alongside DigiSkills and Bano Qabil rather than against them.

DigiSkills and Bano Qabil solve breadth and brand trust at national scale; NeuroNova targets the narrower, specific gap in structured, applied AI/DS depth.

5 Target audience

The primary audience is university students and self-taught learners in smaller Pakistani cities and towns without access to a campus-based AI/DS bootcamp. A secondary audience is university students anywhere in Pakistan who want a structured, practical supplement to coursework that has not caught up to industry needs. The NSCT 2026 assessment recorded rural test participation at less than half the urban rate among IT students, with a marked gender gap in rural districts specifically; the access problem this targets is measurable rather than assumed.³

6 Value proposition

Table 2 sets out what a learner gets from NeuroNova against the two closest comparison points.

7 Basic business and sustainability model

The initial scope is the free, open core, not a fully operating revenue business. Three elements support its sustainability beyond the internship:

- **Community contribution.** Learners who complete the curriculum are invited back as contributors, keeping projects and content current at low ongoing cost.

⁴Superior Group of Colleges, coverage of the Google and Punjab Higher Education Department Career Certificate scholarship partnership, 2026.

Table 2: Value proposition comparison

	NeuroNova	Bano Qabil	Google Career Certificates
Cost	Free	Free	Free (scholarship-gated)
Access	Fully remote	Campus-based cohort	University-channel only
Depth	Structured, project-based AI/DS	Broad digital literacy	Certificate-based
Barrier to start	None	Test, interview, deposit	Enrolment in a partner university

- **Cloud and compute partnerships.** A modest sponsorship for compute credits, similar in spirit to the Alibaba Cloud partnership already running this internship’s AI hackathon.
- **Optional premium layer, later.** A future, clearly separate paid track, such as structured mentorship or portfolio review, once the free core has real usage. This is explicitly not part of the initial scope.

8 Marketing and execution approach

Four channels are already close at hand rather than requiring new infrastructure:

- **GitHub**, the natural discovery channel for a developer-facing, code-first resource.
- **LinkedIn**, for public build-in-progress posts, applying the personal branding practices covered in this internship’s own Week 4 training.
- **University and college outreach**, targeted directly at computer science and STEM societies in smaller cities specifically.
- **Alkhidmat’s own network**, using this internship’s cohort and Volunteer Management contacts as an initial, trusted distribution channel.

9 Key next steps

1. Finish structuring the core project sequence and publish it openly on GitHub.
2. Run a small, informal pilot with a handful of real learners and collect direct feedback, which is the concrete evidence this plan does not yet have (see Section 10).
3. Promote the pilot through GitHub and LinkedIn, and through university outreach in smaller cities.
4. Explore a modest cloud or compute partnership, following the model already in place for this internship’s AI hackathon.

10 Notes on evidence and limitations

This report relies on secondary sources for market sizing and competitor research, cited throughout. It does not yet include primary evidence, such as a short survey or interviews with prospective learners, which the project brief lists as good supporting evidence. A small survey of computer science or IT students, run before the pilot in Section 10's first next step, would be a direct and low-cost way to close this gap and is recommended as a genuine next step rather than treated as already complete.

References

1. Business Recorder, "Pakistan's AI moment," June 2026.
2. Daily Parliament Times, "The Skills Gap Pakistan Cannot Ignore," July 2026.
3. Pakistan Today, HEC National Skills Competition Test (NSCT) 2026 coverage.
4. Superior Group of Colleges, coverage of the Google and Punjab Higher Education Department Career Certificate partnership, 2026.
5. DigiSkills.pk, official website and enrolment pages. <https://digiskills.pk>
6. Bano Qabil, official website and course pages. <https://banoqabil.pk>